Docker lab

1) What is the difference between:

- CMD & ENTRYPOINT

In Docker, the CMD and ENTRYPOINT instructions both define what command runs when a container starts

Cmd: Provides default arguments or a default command.

Entrypoint: Defines the main executable that always runs.

When both ENTRYPOINT and CMD are defined in a Dockerfile, the CMD instruction provides the default arguments to the ENTRYPOINT executable.

- COPY and ADD

The COPY and ADD instructions in a Dockerfile are used to transfer files and directories from the build context (your local machine) into the Docker image

COPY: Copies files/directories as-is from your local context to the image. URL Support, Not supported. Files must be local. Does not extract compressed files copies them as-is. Recommended default for simple file transfers.

ADD: Copies files/directories, and also has extended functionality. URL Support, Can fetch files from remote URLs. utomatically

extracts local compressed files (only local archives, not remote ones). Use only when specific additional features are needed.

Problem 1:

Run the container hello-world

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Check the container status

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED              STATUS
         PORTS     NAMES
8d28a635f341   ubuntu        "bash"     About a minute ago   Exited (0) 47 secon
ds ago             gallant_beaver
c4e189bd9874   hello-world   "/hello"   3 minutes ago        Exited (0) 3 minute
s ago              zealous_cohen
11259d47c744   hello-world   "/hello"   4 minutes ago        Exited (0) 4 minute
s ago              compassionate_khayyam
49e2c9027867   hello-world   "/hello"   20 minutes ago       Exited (0) 20 minut
es ago             focused_wright
```

Start the stopped container

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker start c4e189bd9874
c4e189bd9874
```

Remove the container

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED              STATUS
         PORTS     NAMES
8d28a635f341   ubuntu        "bash"     About a minute ago   Exited (0) 47 secon
ds ago             gallant_beaver
c4e189bd9874   hello-world   "/hello"   3 minutes ago        Exited (0) 3 minute
s ago              zealous_cohen
11259d47c744   hello-world   "/hello"   4 minutes ago        Exited (0) 4 minute
s ago              compassionate_khayyam
49e2c9027867   hello-world   "/hello"   20 minutes ago       Exited (0) 20 minut
es ago             focused_wright
```

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker rm c4e189bd9874
c4e189bd9874
```

Remove the image

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker images
                                                    i Info →   U   In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB    U
ubuntu:latest        d1e2e92c075e        119MB         31.7MB    U
```

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker rmi hello-world
Untagged: hello-world:latest
Deleted: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
albraa@albraa-Cyborg-15-A13UDX:~$
```

Problem 2:

Run container centos or ubuntu in an interactive mode

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
fd8cda969ed2: Download complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
root@8d28a635f341:/# ls
bin   dev  home  lib64  mnt  proc  run   srv  tmp  var
boot  etc  lib   media  opt  root  sbin  sys  usr
```

Run the following command in the container “echo docker ”

Open a bash shell in the container and touch a file named hello-docker

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -it ubuntu bash
root@283727c06b3f:/# echo docker
docker
root@283727c06b3f:/# touch hello-docker
root@283727c06b3f:/# ls
bin   dev  hello-docker  lib    media  opt   root  sbin  sys  usr
boot  etc  home          lib64  mnt    proc  run   srv   tmp  var
root@283727c06b3f:/# ls -l
total 48
lrwxrwxrwx   1 root root    7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root 4096 Apr 22  2024 boot
drwxr-xr-x   5 root root  360 Mar  6 08:23 dev
drwxr-xr-x   1 root root 4096 Mar  6 08:23 etc
-rw-r--r--   1 root root    0 Mar  6 08:23 hello-docker
```

Stop the container and remove it. Write your comment about the file hello-docker

```
exit
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED          STATUS
PORTS     NAMES
283727c06b3f   ubuntu    "bash"    3 minutes ago    Exited (0) 11 seconds ago
          distracted_shirley
8d28a635f341   ubuntu    "bash"    27 minutes ago   Exited (0) 26 minutes ago
          gallant_beaver
albraa@albraa-Cyborg-15-A13UDX:~$ docker stop ^C
albraa@albraa-Cyborg-15-A13UDX:~$ docker stop 283727c06b3f
283727c06b3f
albraa@albraa-Cyborg-15-A13UDX:~$ docker rm 283727c06b3f
283727c06b3f
albraa@albraa-Cyborg-15-A13UDX:~$
```

comment about the file hello-docker

hello-docker file in the container and we remove the container this is mean every thing in container deleted

Remove all stopped containers

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker rm $(docker ps -a -q)
8d28a635f341
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
albraa@albraa-Cyborg-15-A13UDX:~$
```

Problem 3:

Deploy a MySQL database called app-database. Use the mysql latest image, and use the

-e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name app-database -e MYSQL_ROO
T_PASSWORD=P4sSwOrdO! mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
357926362d31: Pull complete
c89a19cdad3b: Pull complete
f3702ac323ca: Pull complete
61941d436e88: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
74a8e4bbd9fe: Pull complete
00af5f73db53: Pull complete
4a629f1008ff: Pull complete
8314ff001dec: Pull complete
b222ef59b124: Download complete
1dc8eadaef55: Download complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
b49c7045b4fbbccf5bd708796b1536c1488e730beeea81413c1cbf1118f456f0
```

-d : run container run in the background.

-name : name for container

-e MYSQL_ROOT_PASSWORD=P4sSwOrdO! : password for root

mysql:latest : pull last version of image mysql

Problem 4:

Run the image Nginx

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name webs -p 8000:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
206356c42440: Pull complete
b47f187216b6: Pull complete
1ad233904a11: Pull complete
df0b66c867e4: Pull complete
eedda9fd8786: Pull complete
17d0911eaf62: Pull complete
35ff83c394d6: Pull complete
0019bbecc572: Download complete
c13150b54a53: Download complete
Digest: sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Status: Downloaded newer image for nginx:latest
afd2554cd7a930a6303a2091f88722427ac6216038b02af4b236a79b0882e870
albraa@albraa-Cyborg-15-A13UDX:~$ docker exec -it webs bash
root@afd2554cd7a9:/# ls
```

Add html static files to the container and make sure they are accessible

```
root@afd2554cd7a9:/# cd /usr/share/nginx/html
root@afd2554cd7a9:/usr/share/nginx/html# echo "<h1> hello albraa from docker ngi
nx </h1>" > index.html
root@afd2554cd7a9:/usr/share/nginx/html# cat index.html
<h1> hello albraa from docker nginx </h1>
root@afd2554cd7a9:/usr/share/nginx/html# exit
exit
albraa@albraa-Cyborg-15-A13UDX:~$
```

hello albraa from docker nginx

Commit the container with image name IMAGE_NAME

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS
        PORTS                                       NAMES
afd2554cd7a9   nginx          "/docker-entrypoint.…"   7 minutes ago    Up 7 min
utes    0.0.0.0:8000->80/tcp, [::]:8000->80/tcp   webs
b49c7045b4fb   mysql:latest   "docker-entrypoint.s…"   17 minutes ago   Up 17 mi
nutes   3306/tcp, 33060/tcp                       app-database
albraa@albraa-Cyborg-15-A13UDX:~$ docker commit webs nginx
sha256:f5a88ed07d30096be7726017803bb0dfe54d44b19eec2f079993e4234e80d696
albraa@albraa-Cyborg-15-A13UDX:~$
```

Problem 5:

Create a python simple app

```
albraa@albraa-Cyborg-15-A13UDX:~$ mkdir python-app
albraa@albraa-Cyborg-15-A13UDX:~$ cd python-app
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ touch app.py
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ vim app.py
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ cat app.py
print("Welcome to My Simple Python App!")
name = input("Enter your name: ")
print(f"Hello, {name}! This is running inside Docker.")
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ touch dockerfile
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ nano dockerfile
```

Create a dockerfile to containerize the python app

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ touch dockerfile
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ nano dockerfile
```

```
  GNU nano 7.2                    dockerfile
FROM python:3.12.3

WORKDIR /app

COPY app.py .

CMD ["python3", "app.py"]
```

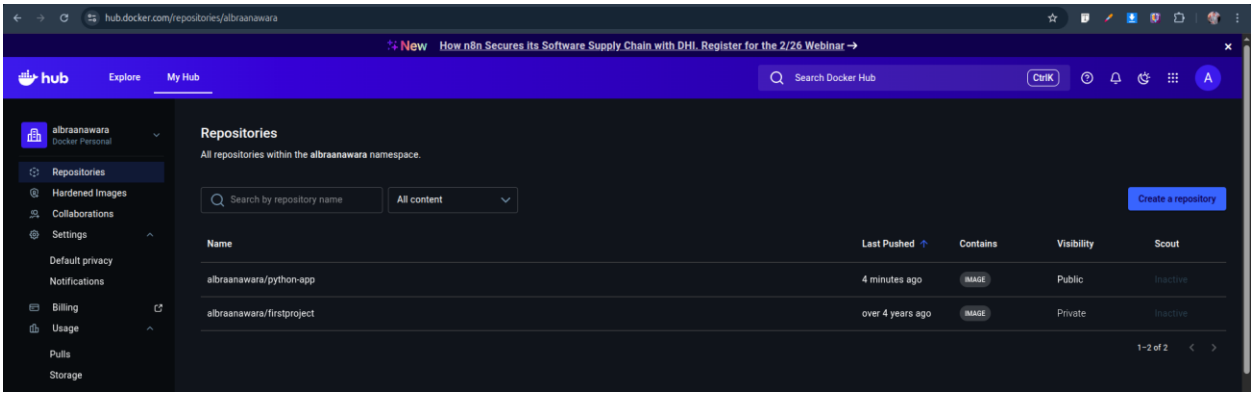
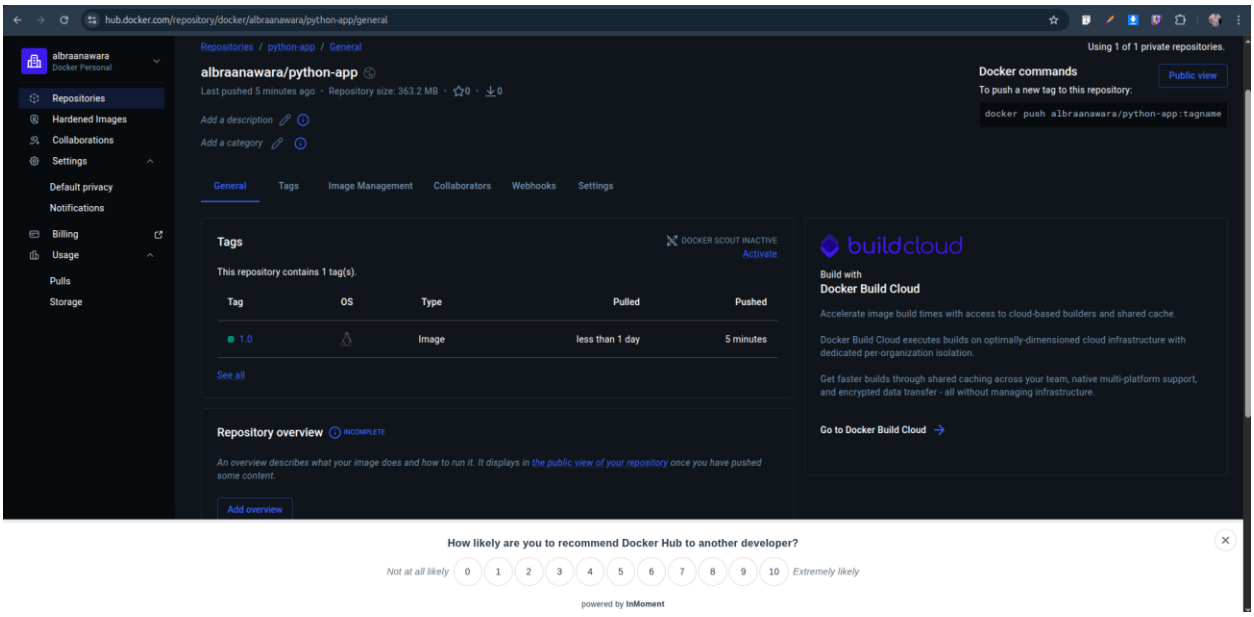
Build the image and test it

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker build -t simple-python-app .
[+] Building 138.6s (8/8) FINISHED                                docker:default
 => [internal] load build definition from dockerfile                        0.0s
 => => transferring dockerfile: 112B                                        0.0s
 => [internal] load metadata for docker.io/library/python:3.12.3            2.1s
 => [internal] load .dockerignore                                           0.0s
 => => transferring context: 2B                                             0.0s
 => [1/3] FROM docker.io/library/python:3.12.3@sha256:3966b81808d864099   135.4s
 => => resolve docker.io/library/python:3.12.3@sha256:3966b81808d864099f8   0.0s
 => => sha256:2a4ca5af09fa6cc704c04ba61e304cbf3d0fb2c55f5 2.77MB / 2.77MB   2.2s
 => => sha256:4429b810e09ef699c3034e49b60f85add3610968983af40 243B / 243B   0.7s
 => => sha256:d46a03def8d9bd846f37fee0edc9fcb7400d6816 22.69MB / 22.69MB   41.9s
 => => sha256:a99509a323905a80628005e4f3bc26ac15ebaf3ffd 6.39MB / 6.39MB   17.7s
 => => sha256:6582c62583ef22717db8d306b1d6a0c288089ff6 64.14MB / 64.14MB   89.2s
 => => sha256:bf2c3e352f3d2eed4eda4feeed44a1022a881 211.21MB / 211.21MB   128.3s
 => => sha256:891494355808bdd3db5552f0d3723fd0fa826675 24.05MB / 24.05MB   55.0s
 => => sha256:c6cf28de8a067787ee0d08f8b01d7f1566a508b5 49.58MB / 49.58MB   49.2s
 => => extracting sha256:c6cf28de8a067787ee0d08f8b01d7f1566a508b56f6e5496   0.9s
 => => extracting sha256:891494355808bdd3db5552f0d3723fd0fa826675f7748537   0.3s
 => => extracting sha256:6582c62583ef22717db8d306b1d6a0c288089ff607d9c0d2   1.1s
 => => extracting sha256:bf2c3e352f3d2eed4eda4feeed44a1022a881058df20ac05   3.8s
 => => extracting sha256:a99509a323905a80628005e4f3bc26ac15ebaf3ffdb08a96   0.2s
 => => extracting sha256:d46a03def8d9bd846f37fee0edc9fcb7400d68166e99f039   0.4s
 => => extracting sha256:4429b810e09ef699c3034e49b60f85add3610968983af403   0.0s
```

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker run -it simple-python-app
Welcome to My Simple Python App!
Enter your name: albraa
Hello, albraa! This is running inside Docker.
```

Push the created image into your docker hub repo

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <
username>'


Your one-time device confirmation code is: CPVW-PBBP
Press ENTER to open your browser or submit your device code here: https://login.
docker.com/activate

Waiting for authentication in the browser…


WARNING! Your credentials are stored unencrypted in '/home/albraa/.docker/config
.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
```

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker tag simple-python-app albraa
nawara/python-app:1.0
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker image ls
                                                          i Info →   U   In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
albraanawara/python-app:1.0    2a4fb1ae4038       1.47GB          381MB    U
mysql:latest                   e5dc14f6e01c       1.27GB          283MB    U
nginx:latest                   f5a88ed07d30        237MB         62.9MB
simple-python-app:latest       2a4fb1ae4038       1.47GB          381MB    U
ubuntu:latest                  d1e2e92c075e        119MB         31.7MB
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker push ^C
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker push albraanawara/python-app
:1.0
The push refers to repository [docker.io/albraanawara/python-app]
b71c6fb30671: Pushed
6582c62583ef: Pushing  62.91MB/64.14MB
d46a03def8d9: Pushed
4429b810e09e: Pushed
2a4ca5af09fa: Pushed
891494355808: Pushed
354a78bbd9b7: Pushed
d848e6e2a986: Pushed
bf2c3e352f3d: Pushing  113.2MB/211.2MB
a99509a32390: Pushed
```

```
albraa@albraa-Cyborg-15-A13UDX:~/python-app$ docker push albraanawara/python-app
:1.0
The push refers to repository [docker.io/albraanawara/python-app]
b71c6fb30671: Pushed
6582c62583ef: Pushed
d46a03def8d9: Pushed
4429b810e09e: Pushed
2a4ca5af09fa: Pushed
891494355808: Pushed
354a78bbd9b7: Pushed
d848e6e2a986: Pushed
bf2c3e352f3d: Pushed
a99509a32390: Pushed
c6cf28de8a06: Pushed
1.0: digest: sha256:2a4fb1ae40380ebc5027f10dfa29fb94267d9a6e9a05e575738b221fd43a
8acb size: 856
albraa@albraa-Cyborg-15-A13UDX:~/python-app$
```